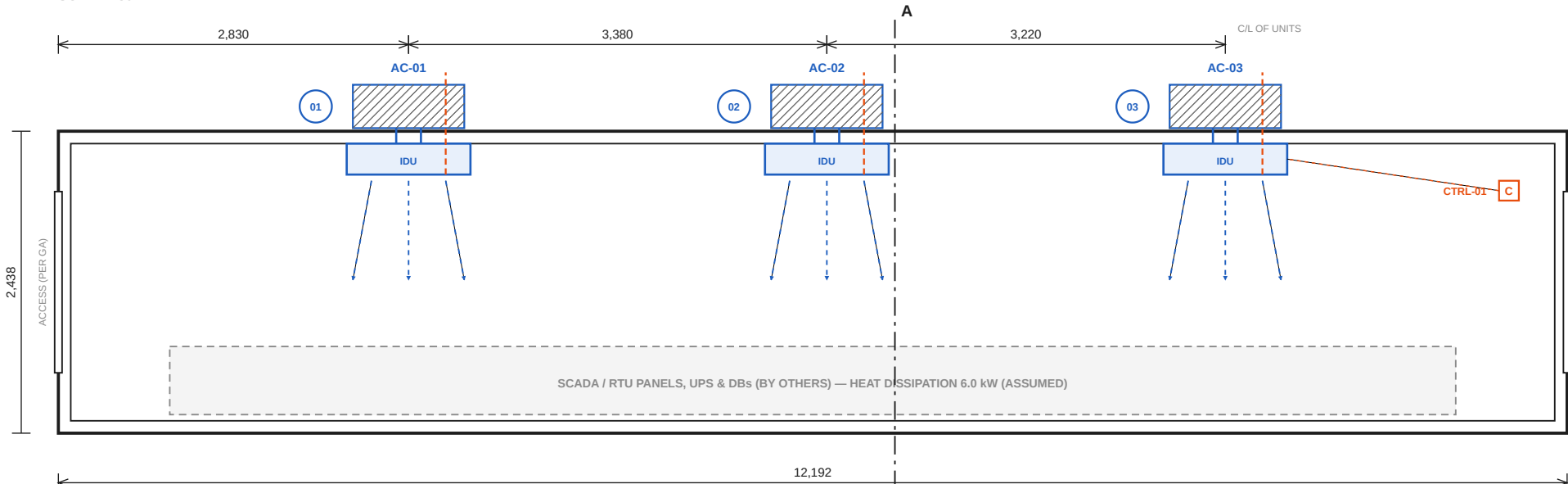


FOR REVIEW — NOT FOR CONSTRUCTION · DESIGN BASIS ASSUMPTIONS TO BE CONFIRMED

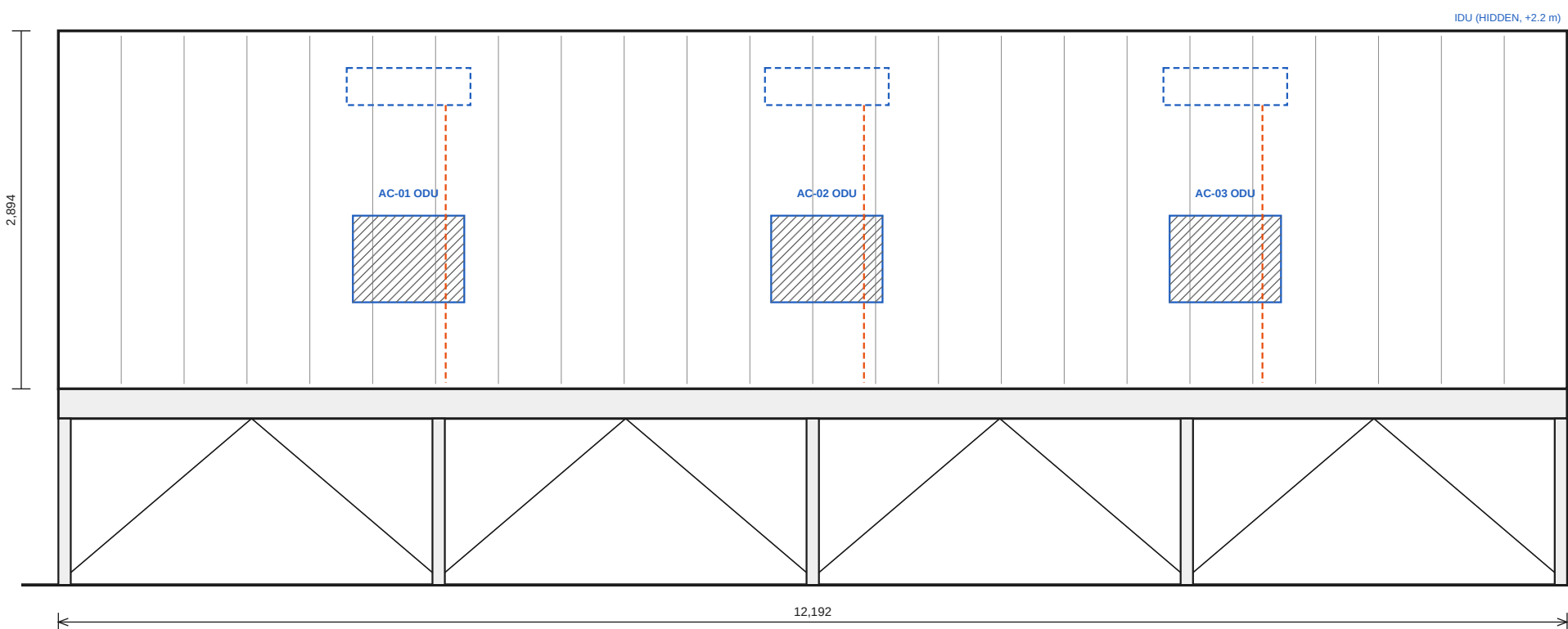
PLAN — SCADA ROOM HVAC LAYOUT

SCALE 1:50



FRONT ELEVATION (A/C WALL) — PER GA SHEET 1

SCALE 1:50

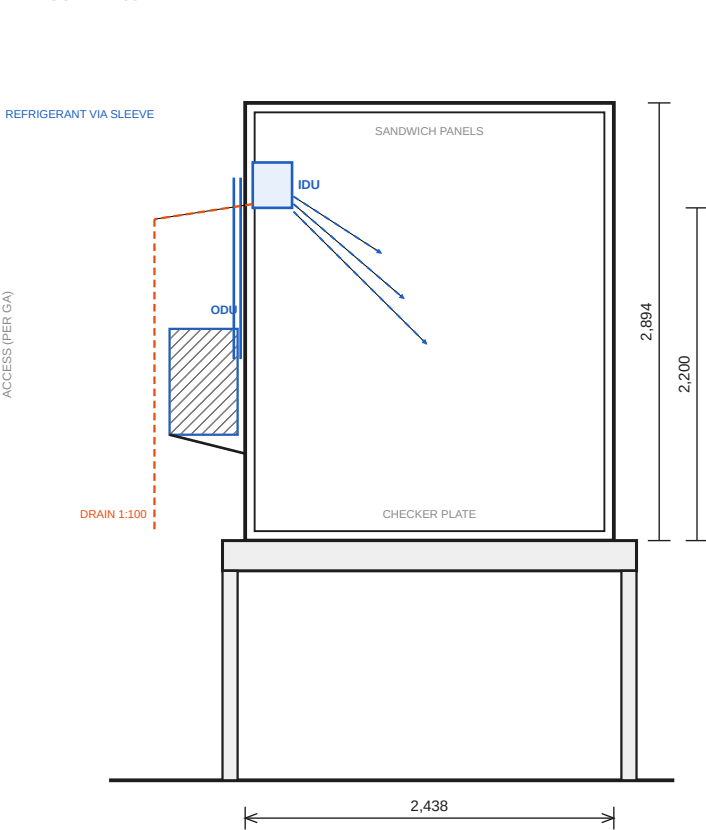


GENERAL NOTES

- All dimensions in mm unless noted. Shelter geometry and A/C positions from GA drawing SWS-2286 Rev 0, sheets 1–2.
- Load calculation: SWS-SCADA-Room-HVAC-Heat-Load-and-Selection.xlsx (simplified CLTD, ASHRAE). Assumptions to be confirmed.
- Units: T3 tropical-rated split A/C, coastal-protected condenser coil, auto-restart, low/high-ambient protection.
- Redundancy N+1: any two units hold 24 °C at design ambient. Controller rotates duty every 24 h, starts standby at 28 °C or on fault.
- Fire & gas: all A/C units tripped by F&G panel on confirmed fire; common alarm and high-temperature alarm to SCADA.
- Condensate: gravity drain, min 1:100 fall, discharged outside clear of skid, cable trays and access platforms.
- Wall penetrations: sealed sleeves, fire-rated where required; ODU guards galvanised mesh per GA.

SECTION A-A

SCALE 1:50



LEGEND

- Split A/C indoor unit (IDU)
- Outdoor unit with mesh guard (ODU)
- Refrigerant pair via sealed sleeve
- Condensate drain (gravity)
- Supply air throw
- Lead/lag controller CTRL-01
- Equipment zone (by others)

EQUIPMENT SCHEDULE

TAG	FROM LEFT END	NOMINAL	TYPE	MODE
AC-01	2.83 m	2.5 TR / 8.8 kW	T3 split, wall IDU	Lead / lag
AC-02	6.21 m	2.5 TR / 8.8 kW	T3 split, wall IDU	Lead / lag
AC-03	9.43 m	2.5 TR / 8.8 kW	T3 split, wall IDU	Lead / lag
CTRL-01	Entrance	—	Duty-standby ctrl	24 h rotation

DESIGN BASIS & LOAD SUMMARY

ITEM	VALUE
Outdoor design	50 °C DB · 22 g/kg
Indoor design	24 °C · ~50 % RH
Envelope U-value	0.45 W/m²K walls & roof
Equipment heat (assumed)	6.0 kW
Design cooling load	12.15 kW (3.45 TR) incl. 10 % margin
Sensible / latent	10.94 kW / 1.21 kW
Selection (N+1)	3 × 2.5 TR · 2 running = 14.1 kW at design ambient

 شركة حلول اللحام المختص للصناعة SPECIALIST WELDING SOLUTIONS CO. FOR INDUSTRIAL Specialist Welding Solutions Co. for Industrial Dammam 2nd Industrial City, KSA INFO@SWSWELD.COM Aramco Approved Vendor 10118949 · CR 2050082045	TITLE	40FT E-HOUSE — SCADA ROOM — HVAC LAYOUT
	CLIENT	(withheld)
	DWG NO	SWS-2286-HVAC-01
	REV	A — FOR REVIEW
	DATE	04/10/2026
	SCALE	AS SHOWN (A3)
	SHEET	1 OF 1